Project G8: CNNs vs Vision Transformers — Pretraining and Data Augmentation on PASCAL VOC

- **Excellent experimental design**: The 2×4×5 factorial structure (pretrained vs scratch × 4 architectures × 5 data fractions) with 3 seeds per configuration is rigorous and well-scoped. The data-scaling analysis (5% to 100%) directly tests the core hypothesis about ViT data efficiency. The literature review demonstrates strong understanding of *why* ViTs need more data (no inductive bias for locality, translation equivariance, or spatial continuity).

- **Multi-label framing is a genuine contribution**: Most CNN-vs-ViT comparisons use single-label datasets (ImageNet, CIFAR). Using PASCAL VOC's multi-label classification (mAP metric) adds a dimension not well-explored in the literature you cite. This is a strength—lean into it.

- **Two concerns**: (1) The proposal lacks an ablation plan for data augmentation despite the title mentioning it—"comparable data augmentation policies" is too vague. Define exactly what augmentations will be used and how you will ablate them. (2) No evaluation beyond mAP is discussed—add per-class AP analysis, confusion patterns, and computational cost comparison.

Recommended actions before Milestone-2

- **Dataset verification**: Download PASCAL VOC 2012 (or 2007+2012 combined). Confirm: 11,530 images (5,717 train, 5,823 val for VOC 2012), 20 classes, multi-label. Report class distribution: some classes (person, car) are much more frequent than others (potted plant, sofa). Compute label density: average number of labels per image (∼1.5), fraction of images with ≥3 labels.

- **Define stratified subsets precisely**: For the 5%, 10%, 20%, 50%, 100% data fractions: stratified sampling must preserve approximate class frequencies across all 20 labels. Since this is multi-label, use iterative stratification (e.g., `scikit-multilearn`'s `IterativeStratification`) rather than standard stratified sampling, which does not handle multi-label. Verify that even the 5% subset (∼286 images) has ≥5 examples per class.

- **ResNet-50 baseline (pretrained, 100% data)**: Load `torchvision.models.resnet50(pretrained=True)`, replace final FC with a 20-class linear layer (no softmax—use sigmoid for multi-label). Loss: Binary Cross-Entropy with logits. Optimizer: AdamW (lr=1e-4 for pretrained backbone, lr=1e-3 for new head—use differential learning rates). Input: 224×224, standard ImageNet normalization. Train 30 epochs, early stopping on val mAP. Target mAP: ∼85–90%.

- **ViT-B/16 baseline (pretrained, 100% data)**: Load from `timm` (`vit_base_patch16_224.augreg_in21k`). Same multi-label head and loss. AdamW (lr=1e-5 for backbone, lr=1e-3 for head—ViTs are sensitive to learning rate, start lower). Target mAP: ∼85–92%.

- **From-scratch baselines**: Train ResNet-50 and ViT-B/16 from random initialization on 100% data. Use higher lr (1e-2 for ResNet, 1e-3 for ViT with warmup). Expect ResNet-50 scratch to achieve mAP ∼60–70%, ViT-B/16 scratch to achieve mAP ∼30–50% (ViT from scratch on 5K images will struggle severely).

- **Define augmentation policy explicitly**: Baseline augmentation: RandomResizedCrop(224), RandomHorizontalFlip, ColorJitter(brightness=0.4, contrast=0.4, saturation=0.4, hue=0.1), Normalize. Strong augmentation: add RandAugment(num_ops=2, magnitude=9) or AutoAugment. Test augmentation: Resize(256) + CenterCrop(224) + Normalize (no randomness).

Ablations

- **Architecture × pretraining × data fraction (core experiment)**: For each of the 4 models × 2 pretraining conditions × 5 data fractions = 40 configurations, train with 3 seeds and report mean±std mAP. Present as a line plot: mAP (y-axis) vs data fraction (x-axis), with separate lines for each model+pretraining combination. This is the centerpiece figure of your project.

- **Data augmentation ablation**: For the 2 best-performing configurations (likely pretrained ResNet-50 and pretrained ViT-B/16), compare: (a) no augmentation (resize + crop only), (b) standard augmentation (flip + color jitter), (c) strong augmentation (RandAugment). Hypothesis: augmentation helps ViT more than ResNet because ViTs overfit faster on small datasets. At 5% data, augmentation may improve ViT mAP by +5–10% but ResNet by only +2–3%.
- **DeiT-S vs ViT-B/16**: DeiT-S uses knowledge distillation from a CNN teacher, which injects CNN-like inductive biases into the transformer. Hypothesis: DeiT-S outperforms ViT-B/16 at low data fractions (5–20%) because the distilled knowledge provides implicit spatial structure, but ViT-B/16 catches up or surpasses at 100%.
- **Per-class AP analysis**: Report AP for all 20 classes separately. Hypothesis: ViT outperforms CNN on "context-dependent" classes (e.g., dining table, which is identified by surrounding objects) because self-attention captures long-range dependencies, while CNN outperforms ViT on "texture-defined" classes (e.g., cat, dog) because convolutions are better at local texture features.
- **Computational efficiency**: Report training time, inference time, peak GPU memory, and parameter count for each architecture. Compute mAP-per-FLOP to identify the most efficient architecture. Hypothesis: ResNet-50 is the most efficient (highest mAP per FLOP), ViT-B/16 is the least efficient but achieves highest absolute mAP when pretrained.

Risks & mitigations

- **Risk**: ViT-B/16 from scratch on 5% of PASCAL VOC ($\sim$286 images) may not train at all—loss may not decrease, mAP may be near random ($\sim$5%)—**Mitigation**: This is actually the point! A near-zero mAP at 5% data for ViT-scratch vs $\sim$40% for ResNet-scratch powerfully demonstrates the data-efficiency gap. Report it as a key finding rather than a failure.
- **Risk**: 40 configurations $\times$ 3 seeds = 120 training runs is computationally expensive, especially for ViT-B/16 (86M params)—**Mitigation**: Prioritize: run pretrained models first (most interesting results), then scratch. Use early stopping aggressively (patience=5). For 5% and 10% data fractions, training should be fast (<10 min per run). Budget compute for the full 100% configurations.
- **Risk**: Multi-label stratification at 5% data may produce subsets where some classes have 0–2 positive examples, making AP for those classes unreliable—**Mitigation**: Report per-class sample counts for each data fraction. Exclude classes with <5 positive examples from the mAP computation at that fraction (report both "full mAP" and "supported-class mAP").

Open questions

- The title mentions "data augmentation" as a primary variable but the Approach section says "comparable data augmentation policies" applied uniformly. Will you actually ablate augmentation as a variable (augmentation on/off, weak/strong), or keep it constant? If constant, the title is misleading—update it.
- Why PASCAL VOC rather than COCO (330K images, 80 classes) or a medical multi-label dataset? VOC is a good choice for studying low-resource settings, but acknowledge that the 20-class, 11K-image setting may not generalize to larger benchmarks. Consider a brief sanity check on CIFAR-10 (single-label, 50K images) to verify your findings transfer.
- How will you handle the multi-label nature in metrics? mAP is standard, but also consider: exact match ratio (fraction of images where all 20 labels are predicted correctly), Hamming loss, and subset accuracy. These give complementary views of multi-label performance.
- ResNet-101 vs ResNet-50: the proposal includes both, but the marginal insight from ResNet-101 is limited (it is just a deeper ResNet). Consider replacing ResNet-101 with ConvNeXt-T (a modernized CNN) to provide a stronger CNN baseline and a more interesting comparison point ("ConvNeXt modernizes CNNs with transformer-era training recipes—does this close the gap to ViTs?").